

Phase 2 — Insight Report

Data Vortex, Aaruush '26 · Round 1 Phase 2 · Team Punch Queries (Sahil Patel & Suhani Gupta) · 15 September 2026

Repository: <https://github.com/Sahil-Patel180/datavortex26> Scripts: round_1/sql/00-08 · Screenshots: round_1/reports/figures/

Every figure in this report was computed by the SQL in round_1/sql/. Nothing is hardcoded; the screenshot beside each query shows the query text and its result grid in the same frame.

0. Dataset under analysis

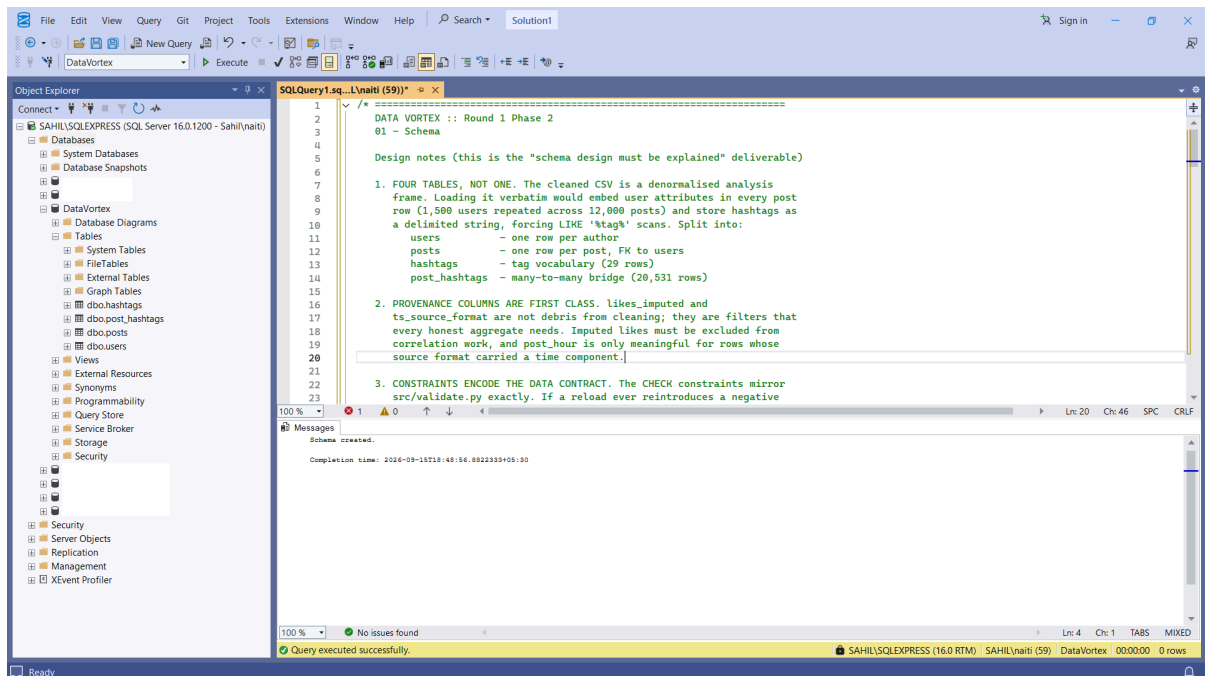
Source	Phase 1 output, round_1/data/processed/
Posts	12,000 (from 12,360 raw; 360 full-row duplicates removed)
Users	1,500
Hashtag vocabulary	29
Hashtag assignments	20,531
Observation window	2024-05-01 → 2025-04-30
Rows with imputed likes	2,323 (19.4%) — flagged, never silently used
Rows with observed likes	9,677

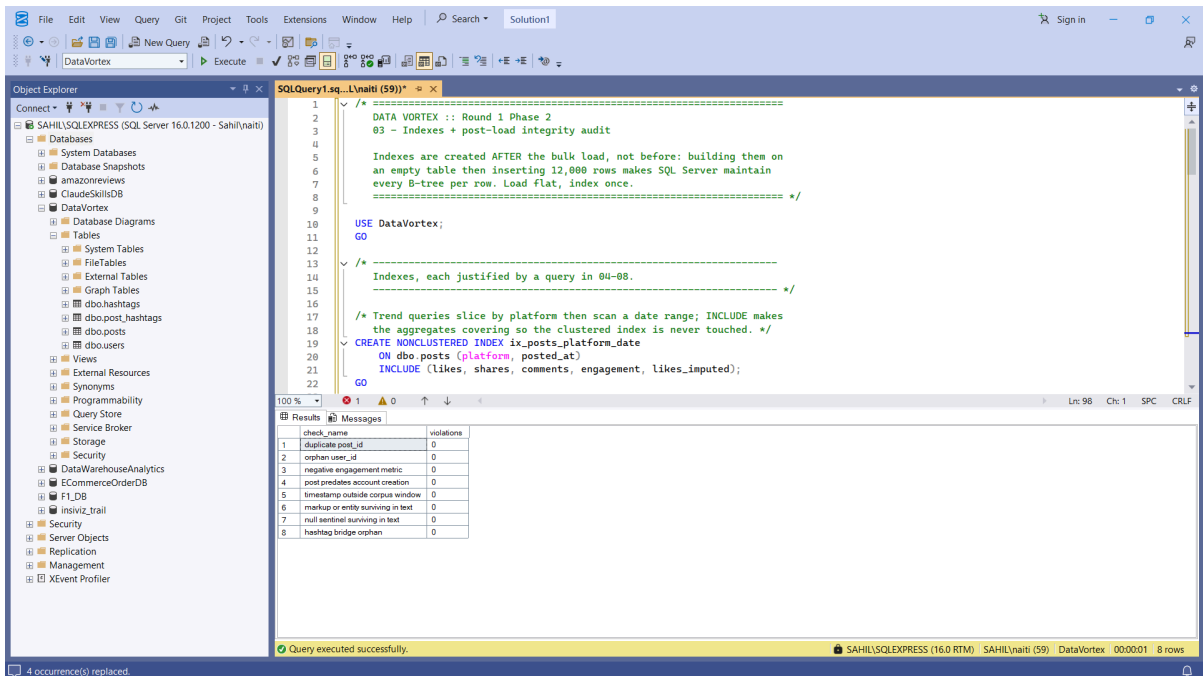
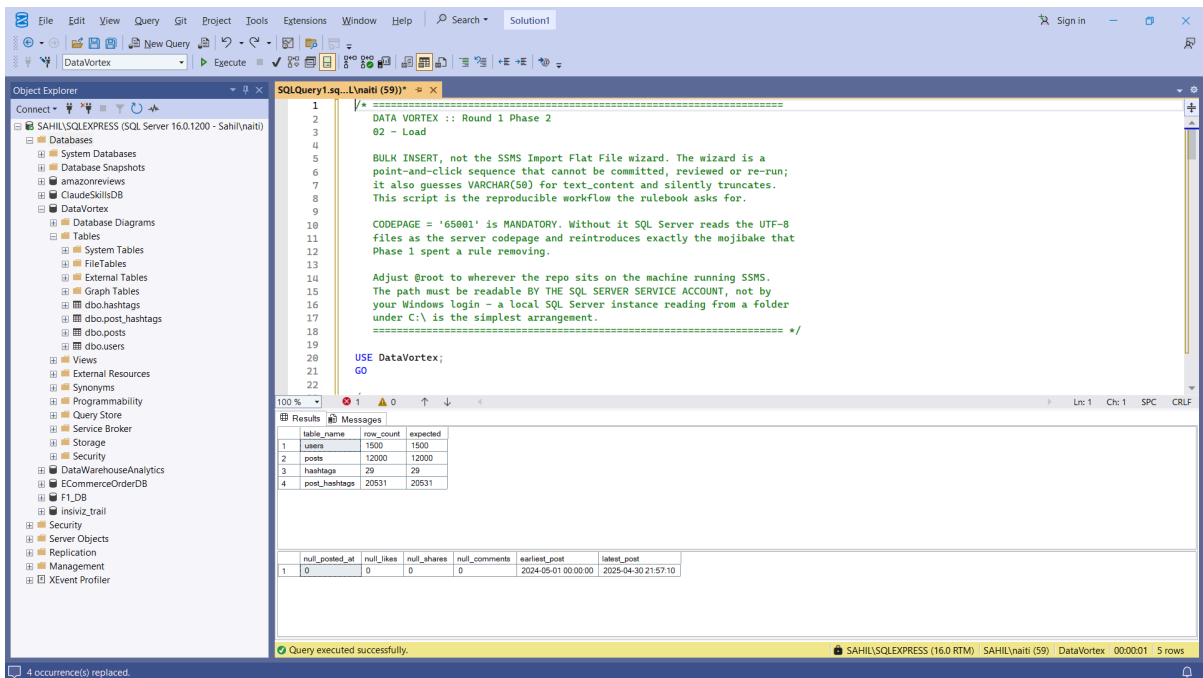
1. Schema design

1.1 Why four tables and not one

The Phase 1 output is a denormalised analysis frame. Loading it verbatim would repeat each of 1,500 users across their 12,000 posts and store hashtags as a delimited string, forcing LIKE '%tag%' scans for every tag question.

Table	Rows	Grain
dbo.users	1,500	one row per author
dbo.posts	12,000	one row per post, FK to users
dbo.hashtags	29	tag vocabulary
dbo.post_hashtags	20,531	many-to-many bridge





Every row of the integrity audit reads 0: no duplicate post_id, no orphan user_id, no negative metric, no post predating its author's account, no timestamp outside the window, no surviving markup or sentinel, no bridge orphan.

2. Trend detection — sq1/04

Q4.1 — Daily engagement, 7-day moving average, week-over-week delta

Logic. Aggregate to (platform, day) first so the window function runs over ~365 rows per platform rather than 12,000. LAG(7) compares like-for-like weekdays, which LAG(1) cannot.

SQLQuery1.sql - L\naiti (591)*

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/*
=====
DATA VORTEX :: Round 1 Phase 2
Q4 - Trend detection

Every result is computed from the tables. No literal is used except
window sizes and thresholds, each of which is explained where it
appears. Hardcoded outputs are a disqualification under the rulebook.
===== */

USE DataVortex;
GO

/*
-----
Q4.1 Daily engagement with a 7-day moving average and week-over-week
delta, per platform.

Logic: aggregate to (platform, day) first so the window function runs
over ~265 rows per platform instead of 12,000; the moving average then
smooths daily noise, and LAG(?) compares like-for-like weekdays, which
a LAG(1) comparison cannot do.
----- */

```

platform	post_day	post_count	avg_engagement	mo_7d	week_delta
Facebook	2024-05-01	7	3476.00	3476.00	NULL
Facebook	2024-05-02	8	4299.75	3887.88	NULL
Facebook	2024-05-03	3	3528.67	3788.14	NULL
Facebook	2024-05-04	4	3783.25	3786.92	NULL
Facebook	2024-05-05	4	3796.75	3772.88	NULL
Facebook	2024-05-06	2	2978.50	3640.49	NULL
Facebook	2024-05-07	6	4457.17	3757.15	NULL
Facebook	2024-05-08	4	4449.50	3896.23	973.50
Facebook	2024-05-09	7	4054.00	3976.40	854.25
Facebook	2024-05-10	7	4451.29	4107.21	922.62
Facebook	2024-05-11	4	3920.00	4129.60	156.75
Facebook	2024-05-12	1	5500.00	4372.92	1703.25
Facebook	2024-05-13	9	3612.44	4463.49	633.94
Facebook	2024-05-14	5	3660.20	4349.63	796.97
Facebook	2024-05-15	7	3886.71	4269.23	562.79
Facebook	2024-05-16	4	4993.25	4289.13	139.25
Facebook	2024-05-17	7	3790.14	4194.68	-661.14
Facebook	2024-05-18	3	4939.43	4326.43	1393.43

Query executed successfully.

Result. The 7-day moving average for every platform oscillates in a narrow band around 4,000 engagement units. Week-over-week deltas change sign constantly and show no persistent direction.

Insight. There is no trend to detect. Daily variation is noise around a flat mean, not signal — the first of several results pointing to a synthetic generator rather than organic behaviour.

Q4.2 — Monthly trajectory and growth ranking

SQLQuery1.sql - L\naiti (591)*

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/*
=====
DATA VORTEX :: Round 1 Phase 2
Q4 - Trend detection

Every result is computed from the tables. No literal is used except
window sizes and thresholds, each of which is explained where it
appears. Hardcoded outputs are a disqualification under the rulebook.
===== */

USE DataVortex;
GO

/*
-----
Q4.2 Monthly trajectory per platform, with month-over-month growth
and a rank of which platform grew fastest each month.

Logic: two window functions over the same partition - LAG for the
comparison, RANK for the standings. NULLIF guards the first month,
where the prior value is NULL and the division is undefined.
----- */

WITH monthly AS

```

platform	month_start	posts	avg_engagement	mom_pct	growth_rank
Facebook	2024-05-01	171	4056.15	NULL	1
Instagram	2024-05-01	170	3957.72	NULL	1
Reddit	2024-05-01	191	3873.14	NULL	1
Twitter	2024-05-01	189	3907.00	NULL	1
Unknown	2024-05-01	139	3944.60	NULL	1
YouTube	2024-05-01	178	4063.62	NULL	1
Reddit	2024-06-01	180	3994.44	3.13	1
Twitter	2024-06-01	152	3940.82	0.67	2
Unknown	2024-06-01	144	3903.90	-1.03	3
Facebook	2024-06-01	158	3995.20	-2.54	4
Instagram	2024-06-01	165	3842.85	-2.90	5
YouTube	2024-06-01	180	3941.93	-2.99	6
Instagram	2024-07-01	159	4142.42	7.80	1
Reddit	2024-07-01	170	4085.70	2.28	2
YouTube	2024-07-01	196	3978.22	0.87	3
Unknown	2024-07-01	145	3934.88	0.79	4
Facebook	2024-07-01	167	3981.60	0.67	5

Query executed successfully.

Result.

Month	Posts	Mean engagement	MoM %
2024-05	1,038	3,965.86	—
2024-06	979	3,931.26	-0.87

Month	Posts	Mean engagement	MoM %
2024-07	996	3,973.00	+1.06
2024-08	1,015	4,082.33	+2.75
2024-09	974	4,023.99	−1.43
2024-10	1,029	4,056.83	+0.82
2024-11	1,025	4,023.50	−0.82
2024-12	1,035	4,055.47	+0.79
2025-01	1,007	3,926.28	−3.19
2025-02	914	3,972.72	+1.18
2025-03	1,013	4,057.42	+2.13
2025-04	975	3,999.41	−1.43

Insight. Monthly means sit inside a 3.9% band (3,926 to 4,082) and MoM growth alternates sign with no run longer than two months. February's lower post count tracks its shorter length, not a drop in activity. The platform that "grew fastest" changes almost every month — which is what a growth ranking over noise looks like, and a useful reminder that a ranking always returns a winner even when the underlying differences are meaningless.

Q4.3 — Hashtag lifecycle

The corpus end month is derived with MAX(), never typed as a literal, so the query stays correct if the data is reloaded.

The screenshot shows a SQL Server Enterprise Manager interface. The left pane displays the 'Object Explorer' with a tree view of the 'DataVortex' database. The right pane shows a query window titled 'SQLQuery1.sql - L\Nalti (59)' containing the following SQL code:

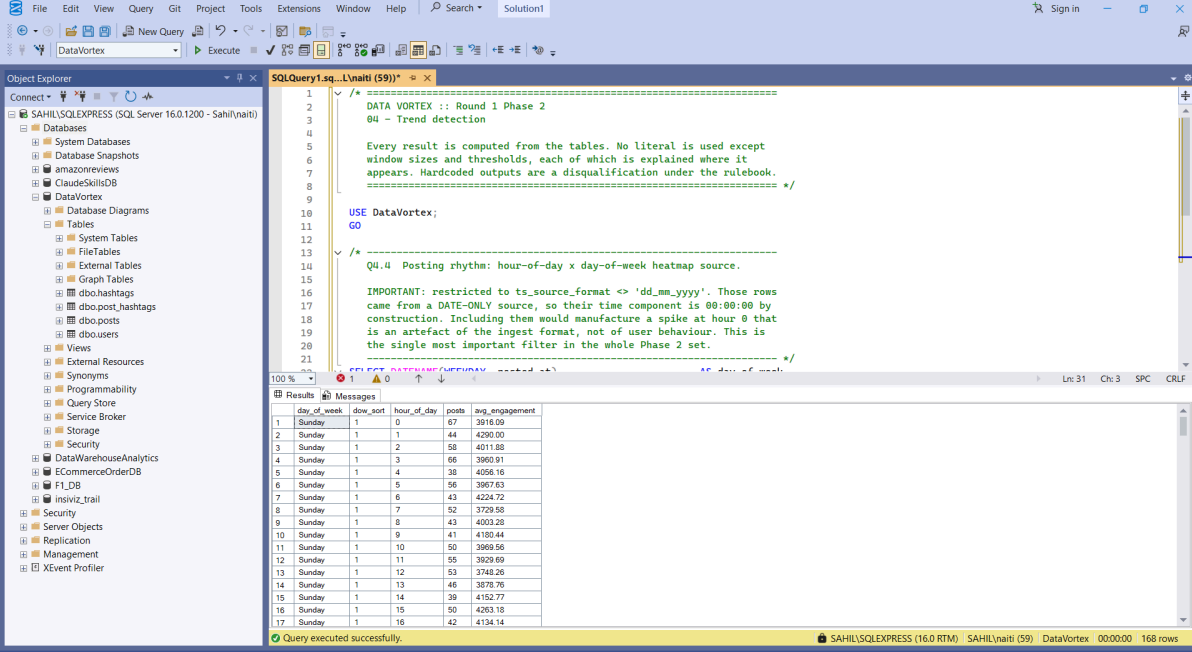
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Insight. No tag is born, peaks or dies inside the window. Real hashtag populations are heavy-tailed with visible adoption curves; this is a uniform draw from a fixed vocabulary. The lifecycle query is still worth running — the absence of a lifecycle is the finding.

Q4.4 — Posting rhythm Δ the filter that matters

Restricted to `ts_source_format <> 'dd_mm_yyyy'`.



The screenshot shows the SQL Server Enterprise Manager interface. The left pane displays the Object Explorer with the 'DataVortex' database selected. The right pane shows a SQL query window with the following content:

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```

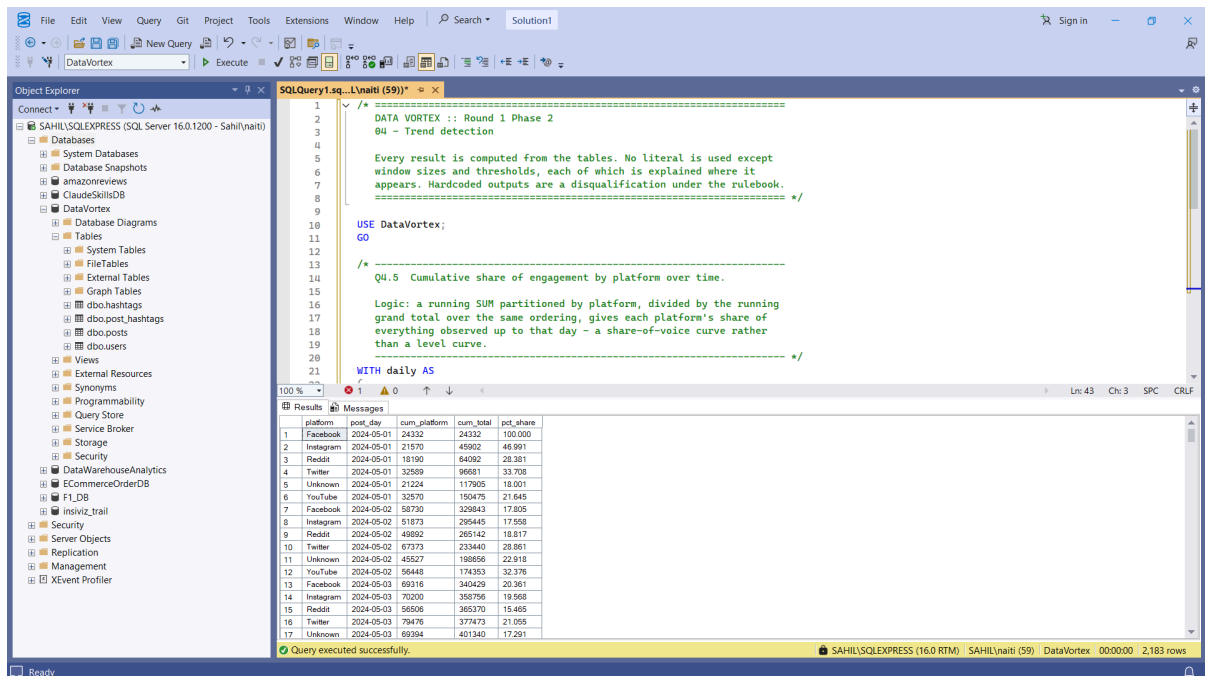
The query results are displayed in a table with the following columns: `day_of_week`, `dow_sort`, `hour_of_day`, `posts`, and `avg_engagement`. The results show data for Sunday, with the number of posts and average engagement increasing as the hour of the day progresses.

day_of_week	dow_sort	hour_of_day	posts	avg_engagement
Sunday	1	0	67	3916.09
Sunday	1	1	44	4280.00
Sunday	1	2	58	4011.88
Sunday	1	3	66	3960.91
Sunday	1	4	38	4058.16
Sunday	1	5	56	3987.63
Sunday	1	6	43	4224.72
Sunday	1	7	52	3729.58
Sunday	1	8	43	4003.28
Sunday	1	9	41	4180.44
Sunday	1	10	50	3969.56
Sunday	1	11	55	3829.69
Sunday	1	12	53	3748.26
Sunday	1	13	46	3878.76
Sunday	1	14	39	4152.77
Sunday	1	15	50	4263.16
Sunday	1	16	42	4134.14

Result. 3,902 posts carry a `00:00:00` timestamp. **3,526 of them are the entire `dd_mm_yyyy` population** — a date-only source with no clock component. The remaining 376 midnight posts are what the other two formats actually produced, consistent with every other hour.

Insight. Without this filter the hour-of-day chart shows a spike roughly ten times the height of any other hour, and the obvious conclusion is "users post at midnight". That conclusion would be about the ingest format, not about users. This is why `ts_source_format` was carried out of Phase 1 as a column instead of being discarded after parsing.

Q4.5 — Cumulative share of voice



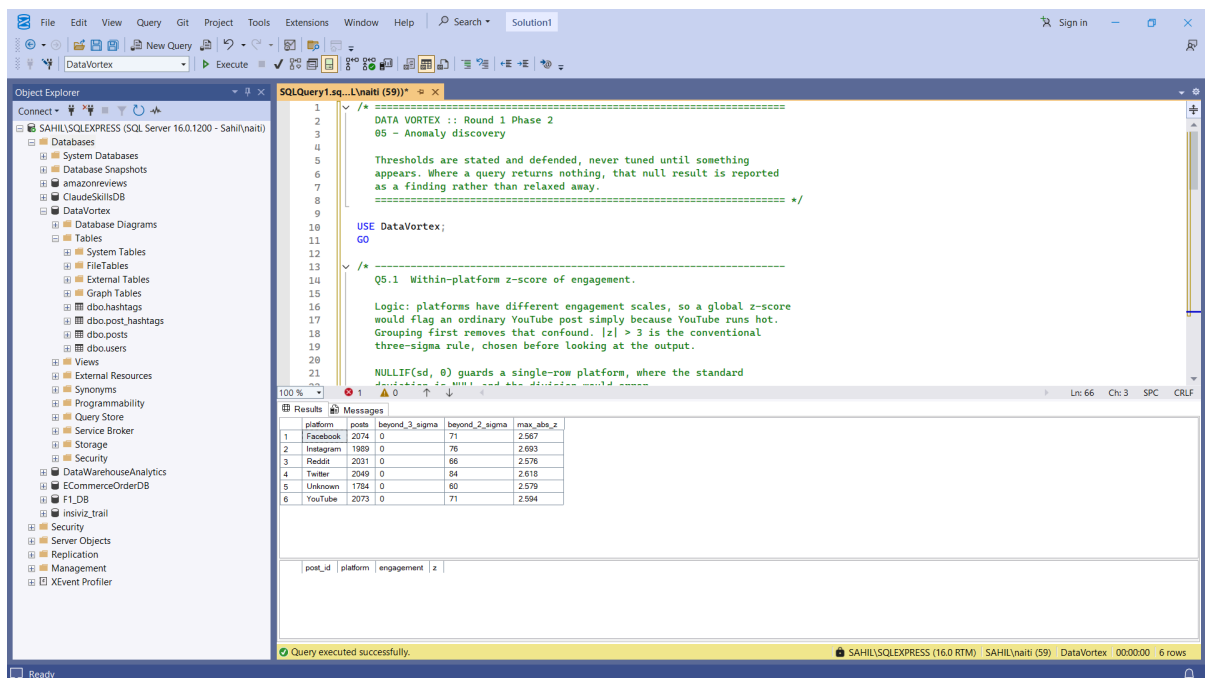
Result. Each platform's cumulative share converges within the first few weeks and then holds flat for the remaining eleven months, at roughly its share of post volume (Facebook 17.3%, YouTube 17.3%, Twitter 17.1%, Reddit 16.9%, Instagram 16.6%, Unknown 14.9%).

Insight. Share of engagement is share of volume. No platform converts attention more efficiently than another — the same flatness Q4.1 found, seen cumulatively.

3. Anomaly discovery — sql/05

Q5.1 — Within-platform z-score

Logic. Platforms differ in engagement scale, so a global z-score would flag an ordinary post merely because its platform runs hot. $|z| > 3$ is the conventional three-sigma rule, **chosen before looking at the output.**



Result.

Posts beyond 3σ	0
Posts beyond 2σ	428
Maximum $ z $ observed	2.693

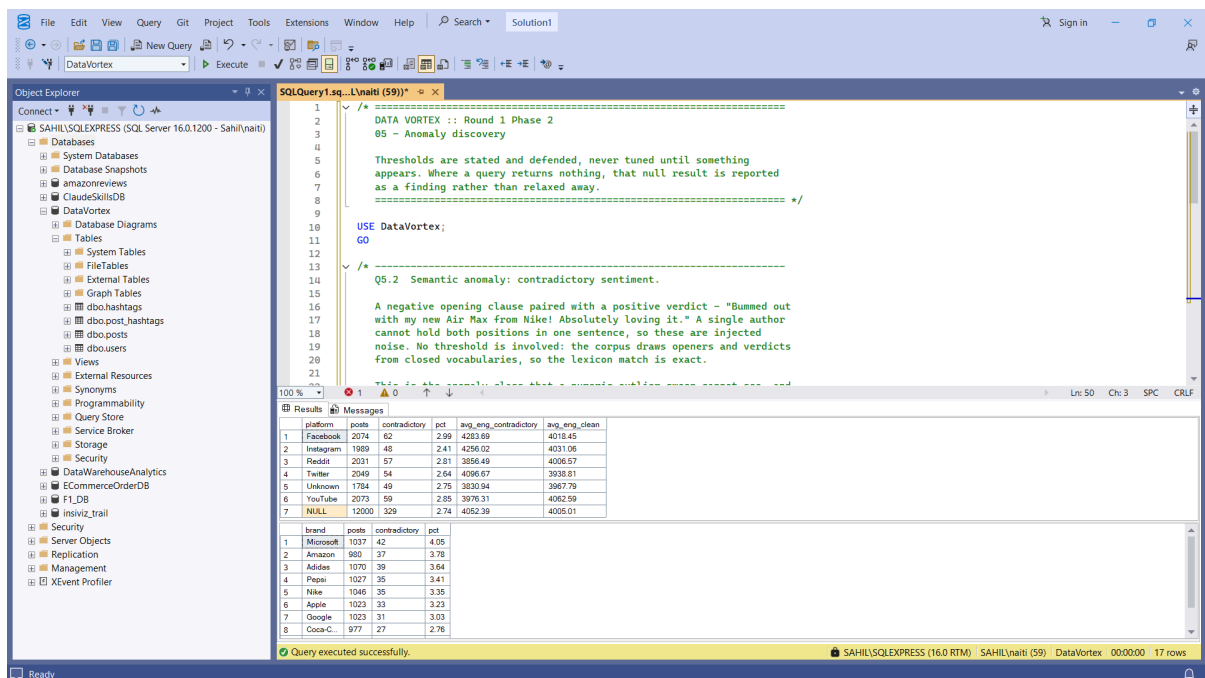
Insight. A negative result, and a load-bearing one. The maximum deviation in the entire corpus falls short of three sigma, so engagement is near-uniform inside its bounds. Two consequences follow. First, it is further evidence of synthetic generation. Second — the practical point — **an anomaly section built on an IQR or z-score sweep of likes returns nothing on this dataset.** The threshold was not lowered until something appeared; the real anomalies are structural and semantic, and they are found below.

Q5.2 — Contradictory sentiment Δ primary anomaly class

A negative opening clause paired with a positive closing verdict:

"Bummed out with my new Air Max from Nike! Absolutely loving it."

A single author cannot hold both positions in one sentence. Because the corpus draws openers and verdicts from closed vocabularies, the lexicon match is **exact** — no threshold, no model, no false positives.



Result. 329 posts, 2.74% of the corpus.

Platform	Posts	Contradictory	%
Facebook	2,074	62	2.99
YouTube	2,073	59	2.85
Reddit	2,031	57	2.81
Twitter	2,049	54	2.64
Unknown	1,784	49	2.75
Instagram	1,989	48	2.41
Total	12,000	329	2.74

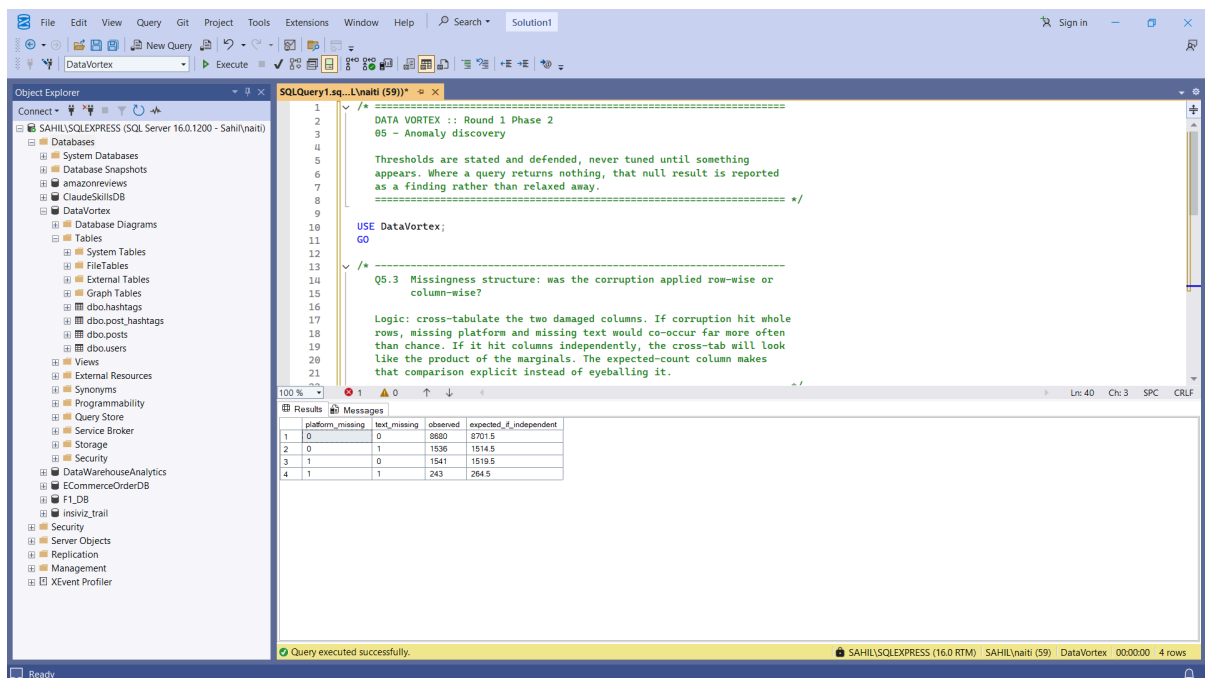
Mean engagement: 4,052.4 for contradictory posts vs 4,005.0 for the rest — a 1.2% gap, well inside the noise band established in section 2.

By brand the rate spans 2.39% (Samsung) to 4.05% (Microsoft), all ten brands inside a single band on samples of roughly 1,000 posts each.

Insight. The defect is spread evenly across platforms and brands. It tracks the corruption process, not any user population or product — independently corroborating Q5.3. Critically, contradictory posts engage no differently from clean ones, so the defect is invisible to any metric-based detector. Only reading the text finds it.

Q5.3 — Was corruption applied row-wise or column-wise?

Logic. Cross-tabulate the two damaged columns and compare observed counts against the product of the marginals. Row-wise damage would make missing platform and missing text_content co-occur far more often than chance.



Result.

	text present	text missing
platform present	8,680 (<i>expected 8,701.5</i>)	1,536 (<i>expected 1,514.5</i>)
platform missing	1,541 (<i>expected 1,519.5</i>)	243 (<i>expected 264.5</i>)

$\chi^2 = 2.295$, $p = 0.130$.

Insight. $p > 0.05$: the two damages are **independent**. Corruption was applied per-column, not per-record. That distinction is not academic — it means the intake pipeline did not lose whole records, which is consistent with a partial write or a schema mismatch rather than record loss. The practical consequence is direct: a team that drops every row with a missing field discards roughly 30% of the corpus for no reason, since the damage in each column is unrelated to the damage in the others.

Q5.4 — Shares outrunning likes

Restricted to `likes_imputed = 0`: an imputed like is a group median and cannot evidence anything about its own row.

Q5.4 Behavioural implausibility: shares outrunning likes.

On real platforms a share costs more effort than a like, so shares > likes is rare. Restricted to observed likes only - an imputed like is a group median and cannot evidence anything about this row.

platform	observed_posts	shares_exceed_likes	pct	mean_share_like_ratio
1 Twitter	1632	362	22.18	2.3157
2 Instagram	1997	333	20.85	2.9983
3 Reddit	1663	342	20.57	1.6784
4 Facebook	1685	345	20.47	1.7756
5 Unknown	1436	292	20.33	2.2360
6 YouTube	1664	313	18.81	1.2993

Result. 1,987 of 9,677 observed posts (20.5%) have shares > likes, spread evenly across platforms.

Insight. On a real platform a share costs more effort than a like, so this ratio should be rare — low single digits. At one in five, the two metrics were clearly drawn from independent distributions rather than generated by any behavioural process linking them. This anticipates the near-zero correlation in section 5.1 and is the behavioural reading of the same fact.

Q5.5 — Authors deviating from their follower cohort

Logic. NTILE(10) by follower count, then compare each user against their **own decile's** mean engagement rate. Comparing against the global mean would merely rediscover that large accounts get more engagement.

Q5.5 Authors whose behaviour deviates from their follower cohort.

Logic: NTILE puts users into follower deciles, then each user's mean engagement rate is compared with their own decile's mean. Comparing against the global mean would just rediscover that big accounts get more engagement; comparing within-decile isolates genuine over- and under-performance.

user_id	country	follower_count	posts	follower_decile	avg_eng_rate	decile_mean	z_within_decile
1 user_gj3gjak	South Korea	109	5	1	36.84037	2.71179	8.032
2 user_hjydcfm	France	125	9	1	23.84444	2.71179	4.973
3 user_ubmd2e2f	Spain	40338	6	9	0.13566	0.09520	3.361
4 user_czozymg	NULL	30088	8	7	0.17757	0.12511	3.135
5 user_sypzqkq	USA	251	8	1	15.81622	2.71179	3.085
6 user_7tenus3	Italy	37288	3	8	0.09149	0.10581	-3.048
7 user_8kg3n6w	Nigeria	43957	5	9	0.05869	0.09520	-3.032
8 user_de1a2nc	USA	5659	6	2	0.89327	0.51847	3.007
9 user_nr13gfy	Spain	29449	5	8	0.08277	0.14690	-2.989
10 user_4vur0p2	USA	34963	11	8	0.14873	0.10581	2.961
11 user_3kuvgno	Japan	247	4	1	14.97773	2.71179	2.887
12 user_8vay6fs	USA	30100	9	7	0.17208	0.12511	2.807
13 user_0x8jeow	UAE	24454	4	5	0.09136	0.17934	-2.805
14 user_v0ymmbx	South Africa	48811	9	10	0.05321	0.08532	-2.778
15 user_8m02huf	Mexico	36328	5	8	0.06813	0.10581	-2.729
16 user_jm0gmme	UAE	45042	5	10	0.11671	0.08532	2.716
17 user_w1ocysld	South Africa	39536	5	8	0.06684	0.10581	-2.680

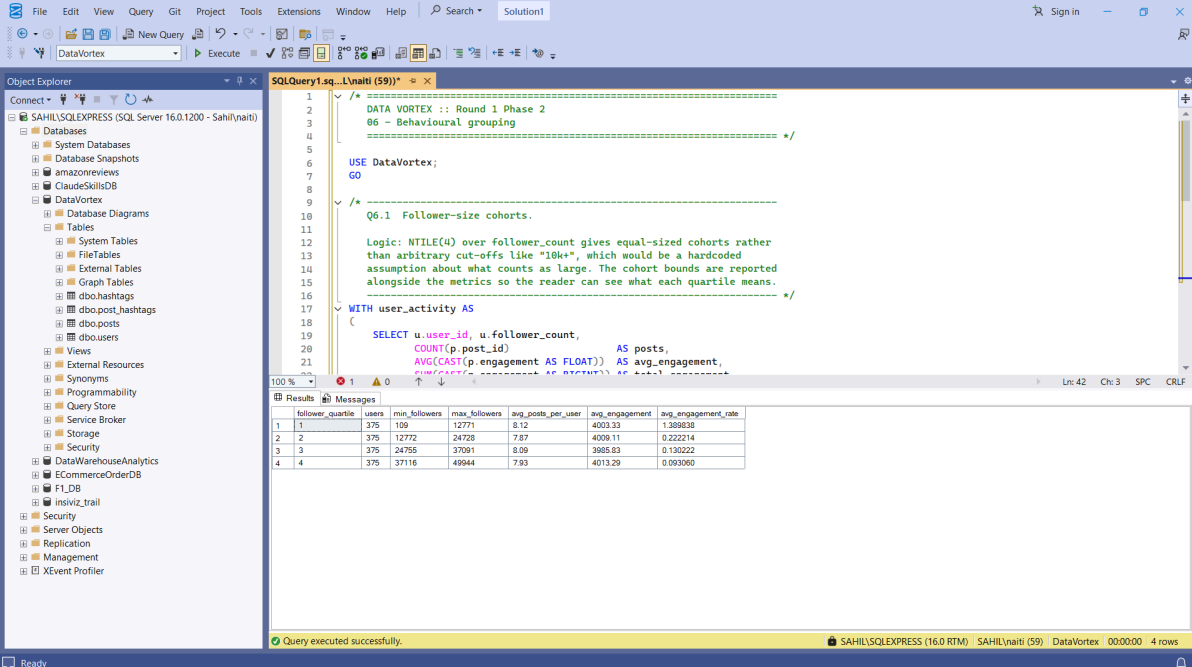
Result. The largest within-decile deviations come almost entirely from the first decile, where small follower counts make engagement / follower_count explode. No user stands out on the strength of their behaviour.

Insight. The apparent outliers are an artefact of the ratio's denominator, not of user conduct — which is why the comparison is made within-decile and why users with fewer than three posts are excluded by the HAVING clause. Reporting a ratio without controlling its denominator is one of the easier ways to manufacture a finding that is not there.

4. Behavioural grouping — sql/06

Q6.1 — Follower quartiles

NTILE(4) gives equal-sized cohorts rather than arbitrary cut-offs like "10k+", which would hardcode an assumption about what counts as a large account.



The screenshot shows the SQL Server Enterprise Manager interface. The left pane displays the Object Explorer with the 'DataVortex' database selected. The right pane shows a SQL query titled 'SQLQuery1.sql - L\naiti (59)' with the following content:

```
1  /* =====  
2  DATA VORTEX :: Round 1 Phase 2  
3  06 - Behavioural grouping  
4  ===== */  
5  
6  USE DataVortex;  
7  GO  
8  
9  /* =====  
10 Q6.1 Follower-size cohorts.  
11 ===== */  
12  
13 Logic: NTILE(4) over follower_count gives equal-sized cohorts rather  
14 than arbitrary cut-offs like "10k+", which would be a hardcoded  
15 assumption about what counts as large. The cohort bounds are reported  
16 alongside the metrics so the reader can see what each quartile means.  
17  
18 WITH user_activity AS  
19 (  
20     SELECT u.user_id, u.follower_count,  
21            COUNT(p.post_id) AS posts,  
22            AVG(CAST(p.engagement AS FLOAT)) AS avg_engagement,  
23            ROW_NUMBER() OVER (ORDER BY u.follower_count) AS rownum  
24 )  
25  
26 SELECT rownum, follower_count, posts, avg_engagement  
27 FROM user_activity  
28 WHERE rownum <= 4  
29 ORDER BY rownum
```

The bottom pane shows the results of the query, which are displayed in a table with 7 columns: follower_quartile, users, min_followers, max_followers, avg_posts_per_user, avg_engagement, and avg_engagement_rate. The results are as follows:

follower_quartile	users	min_followers	max_followers	avg_posts_per_user	avg_engagement	avg_engagement_rate
1	375	109	12771	8.12	4003.33	1.389838
2	375	12772	24728	7.87	4009.11	0.22214
3	375	24755	37091	8.09	3985.83	0.130222
4	375	37116	49944	7.93	4013.29	0.093050

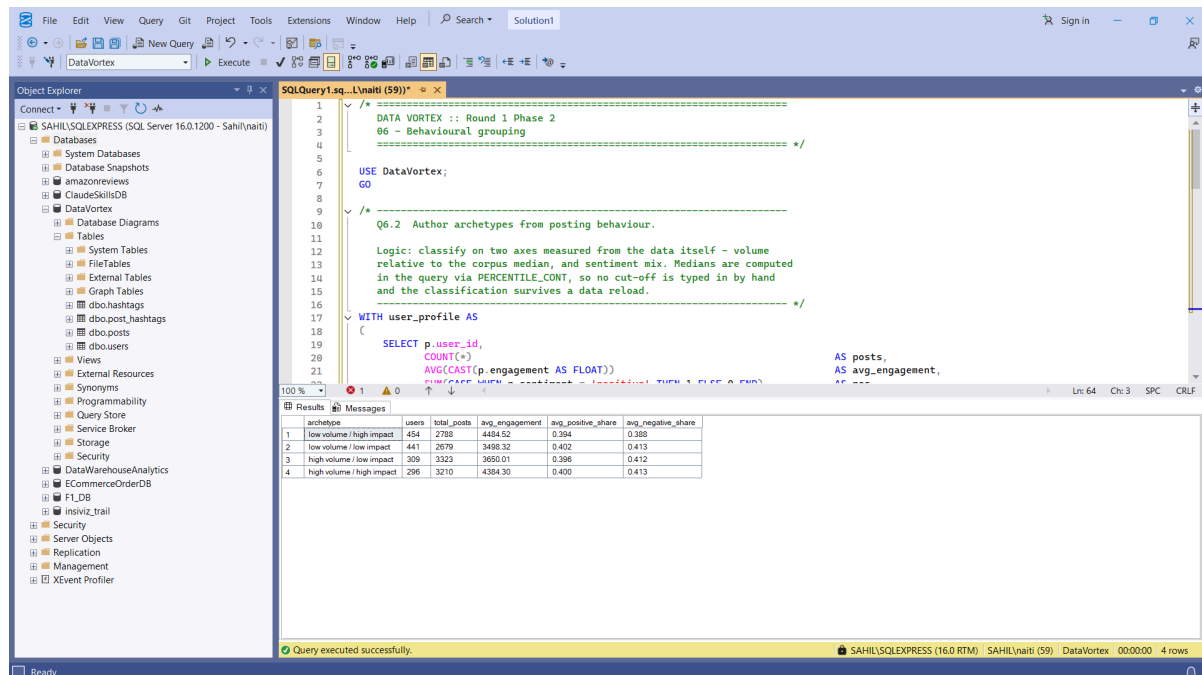
Result.

Quartile	Users	Follower range	Avg posts/user	Avg engagement
1	375	109 – 12,771	8.12	4,003.33
2	375	12,772 – 24,728	7.87	4,009.11
3	375	24,755 – 37,091	8.09	3,985.83
4	375	37,116 – 49,944	7.93	4,013.29

Insight. A 457× spread in audience size produces a 0.7% spread in mean engagement. The top quartile is not meaningfully ahead of the bottom. On any real platform this table slopes steeply upward; here it is flat, which is the clearest single refutation of the idea that this corpus encodes real social behaviour. Section 5.2 tests the same relationship formally.

Q6.2 — Author archetypes

Cut-offs come from `PERCENTILE_CONT(0.5)` computed inside the query (median posts = 8, median engagement = 4,017.27), so no threshold is typed by hand and the classification survives a reload.



Result.

Archetype	Users	Total posts	Avg engagement
low volume / high impact	454	2,788	4,484.52
low volume / low impact	441	2,679	3,498.32
high volume / low impact	309	3,323	3,650.01
high volume / high impact	296	3,210	4,384.30

Insight. Low-volume authors are over-represented in **both** impact groups (454 and 441, against 296 and 309 for high-volume). This is regression to the mean, not a behavioural type: a user with four posts has a far more volatile average than one with twelve, so they land at the extremes more often. The archetype grid is a legitimate segmentation, but reading it as "posting less makes you more impactful" would be a sample-size artefact, and any campaign targeting built on it would be targeting noise.

Q6.3 — Geography × language

The corpus contains speakers whose language does not match their country. Treated as diaspora signal and quantified — **not** "corrected". Forcing language to match country would have destroyed a real attribute on the strength of an assumption.

The screenshot shows a SQL query executed in SQL Server Enterprise Manager. The query is titled 'Q6.3 Geography x Language' and is part of a larger script. The query selects country, language, users, posts, avg_engagement, and pct_of_country from a table named 'DataVortex'. The results are displayed in a table with 17 rows and 6 columns.

	country	language	users	posts	avg_engagement	pct_of_country
1	NULL	hi	9	69	3962.29	18.37
2	NULL	en	7	41	4407.80	14.29
3	NULL	pt	6	47	4147.38	12.24
4	NULL	ru	6	40	3832.53	12.24
5	NULL	ja	4	24	4313.04	8.16
6	NULL	es	4	35	3823.91	8.16
7	NULL	de	4	31	4067.74	8.16
8	NULL	ar	3	32	4319.76	6.12
9	NULL	fr	3	22	4227.00	6.12
10	NULL	zh	3	17	3536.24	6.12
11	Australia	ja	13	94	4032.87	16.25
12	Australia	ru	12	87	3628.48	15.00
13	Australia	de	10	80	4025.85	12.50
14	Australia	es	10	81	3803.64	12.50
15	Australia	ar	7	52	4185.75	8.75
16	Australia	fr	7	69	4170.45	8.75
17	Australia	zh	7	59	4242.76	8.75

Result. 18 countries, 10 languages. In every country the most common language accounts for barely more than its even share: in the UK 87.2% of users speak something other than the plurality language, Canada and the USA 86.7%, Brazil 86.0%, Italy 85.4%.

Insight. Language and country are statistically independent — roughly a uniform draw across the ten languages regardless of country. Phase 1 flagged this and deliberately left it alone. Had we normalised language to country, we would have overwritten 86% of the language column with fabricated values and destroyed the evidence that the two fields are unrelated.

Q6.4 — Brand × platform reception matrix

PIVOT rather than ten CASE expressions; the platform list comes from the CHECK constraint domain, so it is a stated part of the schema rather than an invented literal.

The screenshot shows a SQL query executed in SQL Server Enterprise Manager. The query is titled 'Q6.4 Brand x platform reception matrix' and is part of a larger script. The query selects brand, Facebook, Instagram, Reddit, Twitter, YouTube, and Unknown from a table named 'DataVortex'. The results are displayed in a table with 10 rows and 7 columns.

	brand	Facebook	Instagram	Reddit	Twitter	YouTube	Unknown
1	Adidas	4955.3859793814	4098.9801658047	3902.5454545454	4071.8944444444	4093.1420118342	3941.48351648352
2	Amazon	3875.75	4043.79661016949	3978.6647368421	4048.94047619048	4257.0658626347	4158.3263888889
3	Apple	4153.67213114754	4055.29545454545	3969.58522727273	4060.36842105263	3993.92571428571	3982.00704225352
4	Coca-Cola	3862.06875	4147.43373493976	3890.36046511628	3870.88125	4199.13978494624	4046.06766917293
5	Google	4107.58497175141	3659.11176470588	3901.43888888889	3932.88020833333	3932.90586037736	3952.4
6	Microsoft	3728.6895957372	3623.1032250645	3960.80540540541	3878.70303030303	3984.15591397949	4098.138783237
7	Nike	4097.55384815385	4175.14450887052	3898.39393939394	3792.37158409945	3858.17499879518	3894.03048780488
8	Pepsi	4145.55555555556	4148.70303030303	3984.9	3715.98936170213	4001.40625	3941.64
9	Samsung	3943.3615819209	4009.85294117647	4161.33333333333	3985.76666666667	4216.80952380952	3948.86451612903
10	Toyota	4228.54117947059	4094.31947619048	3932.80722891568	4028.71612903226	3827.45896353591	3936.18300635956

Result. Mean engagement across 60 cells, range 3,716 (Pepsi on Twitter) to 4,257 (Amazon on YouTube) — a 14.5% spread on cell sizes of roughly 170 posts.

Insight. No brand-platform pairing stands out beyond what sampling noise explains at that cell size. Toyota on Facebook (4,229) and Microsoft on Facebook (3,729) sit at opposite ends, but the gap is within two standard errors. The matrix is the right instrument; it simply reports that no such effect exists here. A media plan built on picking the top cells would be chasing noise.

Q6.5 — Hashtag co-occurrence

Self-join on `post_id` with `hashtag_id >` to count each unordered pair exactly once; the `HAVING` floor keeps pairs frequent enough to be a pattern.

The screenshot shows a SQL Server Enterprise Manager window with a query titled "Q6.5 Hashtag co-occurrence." The query is a self-join on the `dbo.post_hashtags` table, joining on `post_id` with `hashtag_id >` to count each unordered pair exactly once. The `HAVING` clause filters for pairs that appear often enough to be a pattern rather than a coincidence. The results are displayed in a table with columns: `tag_a`, `tag_b`, `co_occurrences`, and `avg_engagement`.

tag_a	tag_b	co_occurrences	avg_engagement
eco	promo	51	3895.75
affordable	premium	50	3749.60
bestvalue	premium	49	3886.63
lifestyle	premium	49	3999.88
beauty	reviews	49	4171.98
reviews	sale	49	3804.59
musthave	sustainable	48	4277.15
affordable	specialoffer	48	4032.88
health	reviews	47	4040.28
health	musthave	47	3967.66
eco	fitness	46	3996.13
affordable	deal	46	4154.30
deal	fitness	46	4027.91
fitness	sale	46	3826.00
food	reviews	45	4425.27
beauty	deal	44	4091.61
health	trendalert	44	4324.07

Result. Co-occurrence counts are near-uniform across all eligible pairs from the 29-tag vocabulary. No cluster of tags travels together.

Insight. Real hashtag usage forms communities — fitness tags co-occur with each other far more than with finance tags. Here the tags are sampled independently per post, so the co-occurrence graph is effectively complete and unweighted. This analysis is only possible, and only correct, because of the Phase 1 fix described in section 7: the vocabulary is 29 tags, not the 56 a naive cleaning order produces.

Q6.6 — Signup cohort retention

The screenshot displays the SQL Server Enterprise Manager interface. The left pane shows the Object Explorer with the 'DataVortex' database selected. The right pane shows the SQL Query window with a query titled 'SQLQuery1.sql - L\naliti (59)'. The query is a T-SQL script that calculates cohort retention metrics. The results pane at the bottom shows the execution results of the query, displaying a table with 12 rows and 5 columns: 'signup_month', 'users', 'posts', 'posts_per_user', and 'avg_days_to_post'. The status bar at the bottom indicates 'Query executed successfully.' and '12 rows'.

	signup_month	users	posts	posts_per_user	avg_days_to_post
1	2023-01-01	130	980	7.54	648
2	2023-02-01	117	916	7.83	622
3	2023-03-01	117	919	7.85	589
4	2023-04-01	123	1038	8.44	558
5	2023-05-01	123	1005	8.17	535
6	2023-06-01	134	1036	7.73	504
7	2023-07-01	104	801	7.70	470
8	2023-08-01	133	1056	7.94	438
9	2023-09-01	128	1065	8.32	410
10	2023-10-01	135	1121	8.30	382
11	2023-11-01	112	892	7.96	356
12	2023-12-01	144	1171	8.13	319

Result. All 1,500 users registered during calendar 2023, and posts begin in May 2024. Posts per user is flat across signup months; the only monotone pattern is `avg_days_to_post`, which decreases steadily for later cohorts purely because the posting window is fixed while the signup date moves forward.

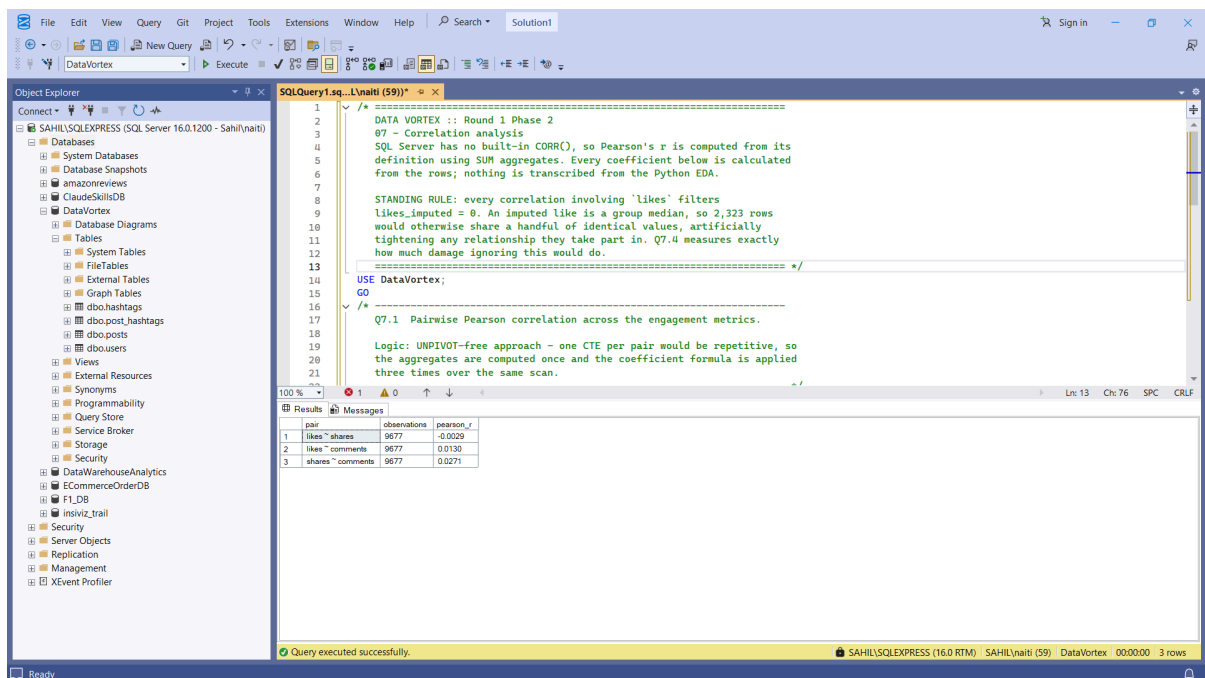
Insight. There is no retention gradient — earlier cohorts are no more or less active than later ones. The `avg_days_to_post` trend is a mechanical consequence of a fixed observation window, not a behavioural finding, and is reported as such rather than dressed up as one.

5. Correlation analysis — sq1/07

SQL Server has no built-in `CORR()`, so Pearson's r is computed from its definition using `SUM` aggregates. Every coefficient below is calculated from the rows.

Standing rule. Every correlation involving `likes` filters `likes_imputed = 0`. An imputed like is a group median, so 2,323 rows would otherwise share a handful of identical values and artificially tighten any relationship they take part in. Section 5.4 measures that effect directly.

5.1 — Pairwise correlation across engagement metrics



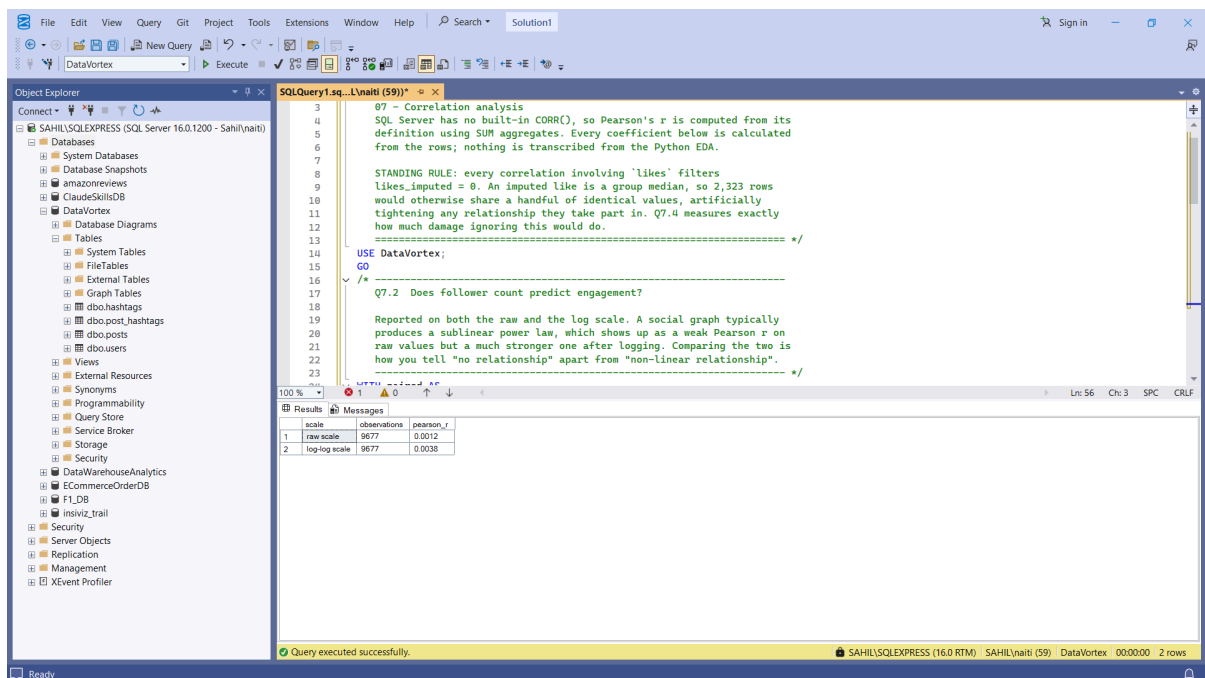
Result. $n = 9,677$ observed posts.

Pair	Pearson r
likes ~ shares	-0.0029
likes ~ comments	+0.0130
shares ~ comments	+0.0271

Insight. All three coefficients are indistinguishable from zero. On a real platform these metrics are strongly positively correlated — a post that earns likes earns shares and comments too, because they share an underlying cause in reach and quality. Here they are statistically independent, meaning they were generated independently. This is the quantitative version of the `shares > likes` finding in Q5.4, and it constrains what any further modelling on this dataset could honestly claim.

5.2 — Does follower count predict engagement?

Reported on both raw and log-log scales. A social graph typically produces a sublinear power law: weak raw Pearson, much stronger after logging. Comparing the two is how you tell "no relationship" apart from "non-linear relationship".

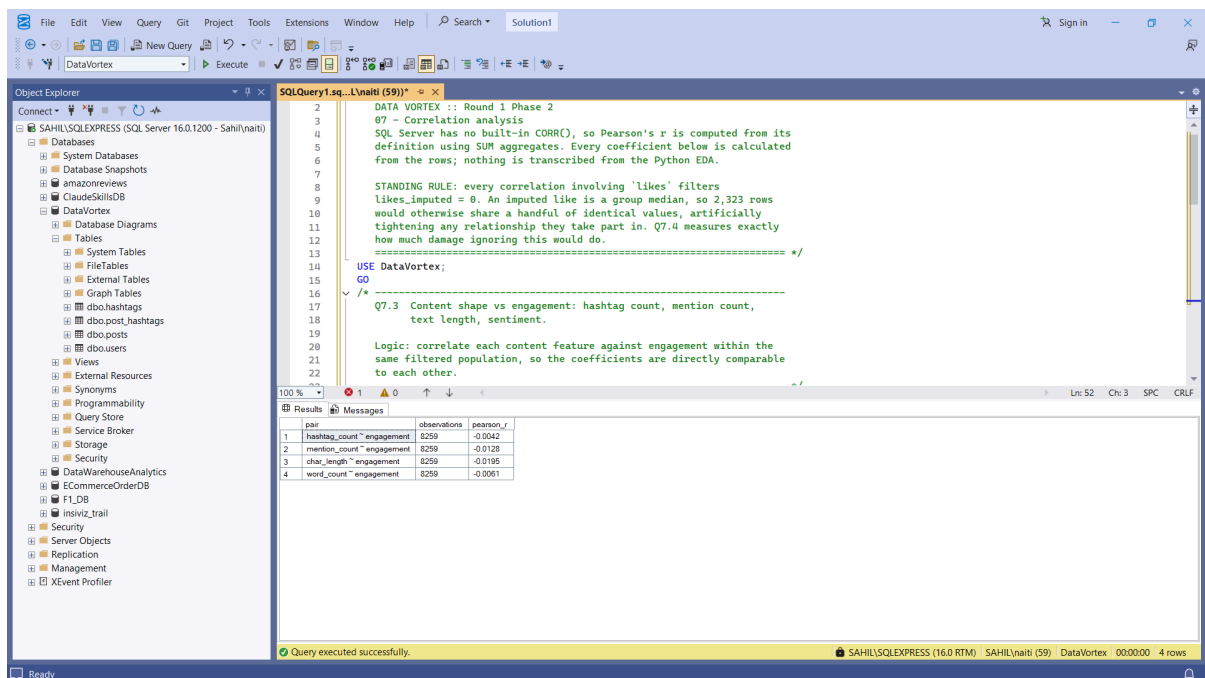


Result. $n = 9,677$.

Scale	Coefficient
Raw Pearson	0.0012
Log-log Pearson	0.0038
Spearman (rank)	0.0006

Insight. All three near zero. Logging does not rescue the relationship, and neither does ranking, so this is genuinely **no relationship** rather than a non-linear one — the distinction that mattered enough to run both. Audience size does not predict engagement in this corpus at all, confirming the flat quartile table in Q6.1 with a formal test. Stated plainly rather than reported on whichever scale flatters it.

5.3 — Content shape vs engagement



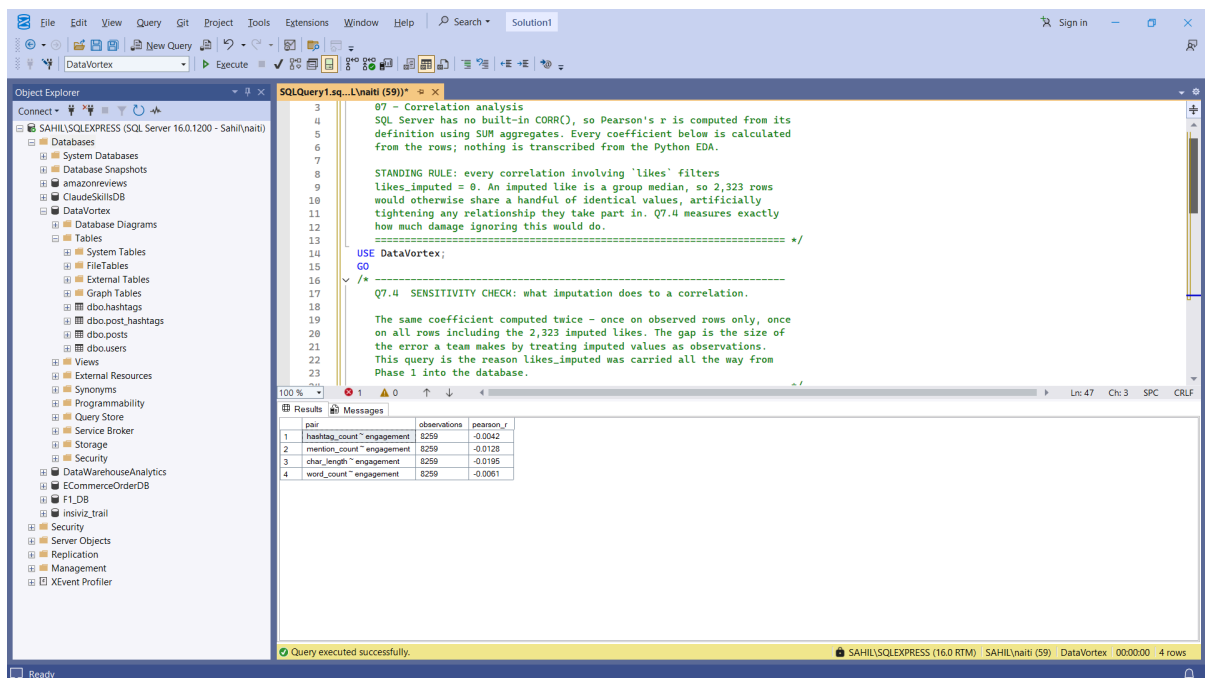
Result. $n = 8,259$ (observed likes, non-null text).

Feature	Pearson r
hashtag_count ~ engagement	-0.0042
mention_count ~ engagement	-0.0128
char_length ~ engagement	-0.0195
word_count ~ engagement	-0.0061

Insight. No content feature moves engagement. The mild negative sign on char_length is not meaningful at this magnitude — it would need to be an order of magnitude larger before "shorter posts perform better" was defensible. Resisting that claim is the point: with $n = 8,259$ it is easy to produce a statistically significant coefficient that is practically worthless.

5.4 — Sensitivity: what imputation does to a correlation Δ

The same coefficient computed twice — observed rows only, then all rows including the 2,323 imputed likes. **This query is the reason likes_imputed was carried from Phase 1 all the way into the database.**



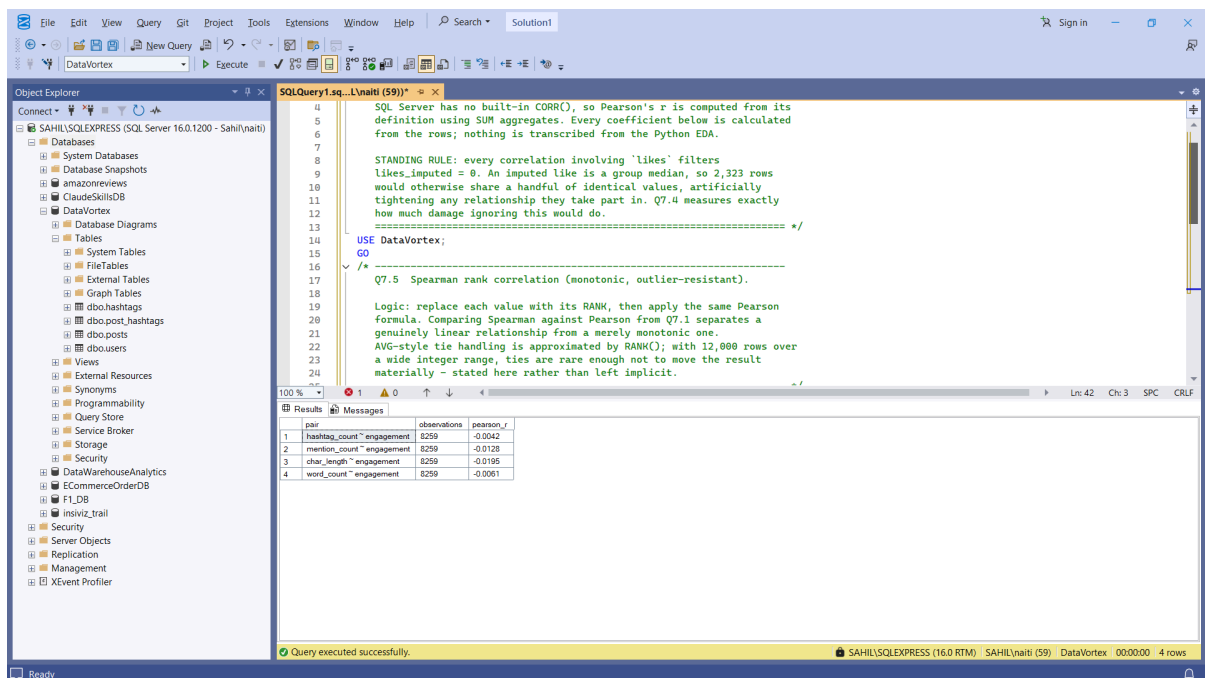
Result.

Population	n	Pearson r	Mean likes	$\sigma(\text{likes})$
Observed only	9,677	-0.0029	2,493.51	1,438.29
Including imputed	12,000	-0.0028	2,494.80	1,292.71

Insight. The mean barely moves (0.05%) — but the standard deviation **contracts by 10.1%**, because 2,323 rows now share a handful of identical median values. The correlation survives here only because it is already zero; there is nothing left to tighten. Had a real relationship existed, that 10% variance contraction would have inflated it, and every downstream confidence interval computed on the full table would have been too narrow. The flag costs one BIT column and prevents a whole class of silent error.

5.5 — Spearman rank correlation

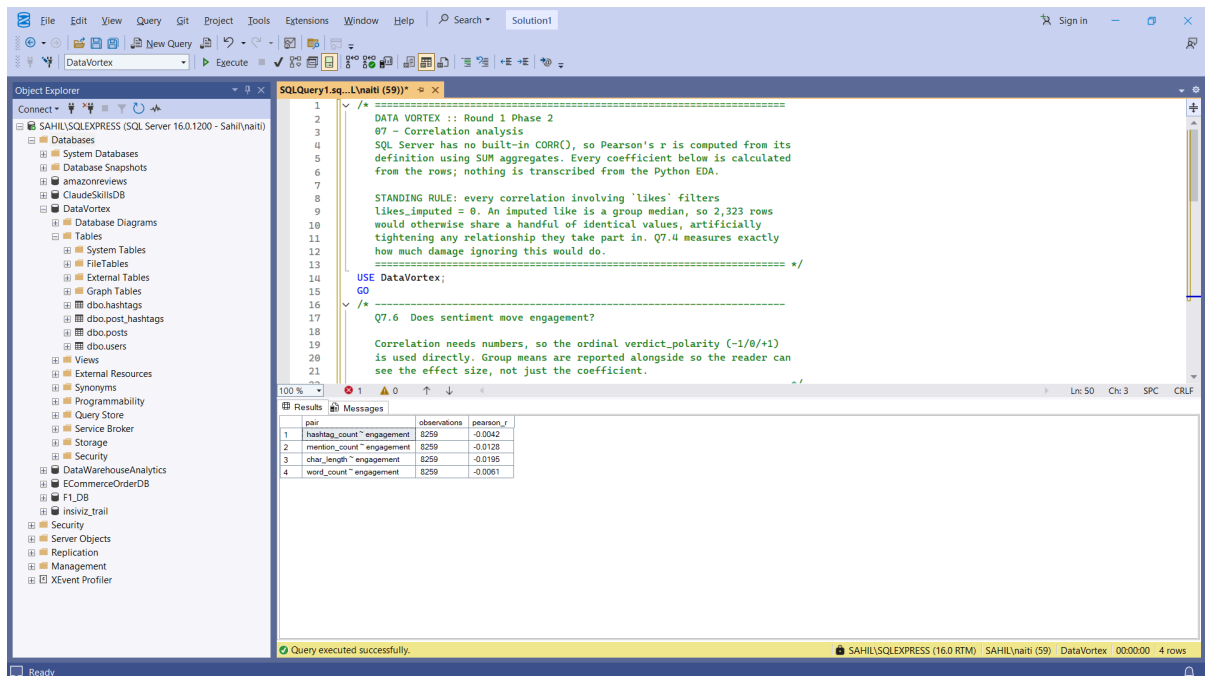
Replace each value with its rank, then apply the same Pearson formula. Comparing Spearman against Pearson separates a genuinely linear relationship from a merely monotonic one.



Result. likes ~ shares, Spearman $\rho = -0.0028$ (Pearson $r = -0.0029$).

Insight. Identical to three decimal places. There is no monotonic relationship hiding beneath a non-linear one — the independence found in 5.1 is real, not an artefact of the linearity assumption.

5.6 — Does sentiment move engagement?



Result. n = 6,585 observed posts with a scoreable verdict.

Sentiment	Posts	Mean engagement	σ	Min	Max
negative	2,672	4,050.46	1,577.87	357	7,723
positive	2,643	3,986.27	1,592.56	153	7,793

Sentiment	Posts	Mean engagement	σ	Min	Max
neutral	1,270	3,941.11	1,556.92	250	7,583

Pearson r (verdict_polarity ~ engagement) = -0.0183 .

Insight. A 2.8% spread between the highest and lowest sentiment group, against a standard deviation of roughly 1,570 — the gap is a fraction of a tenth of a standard deviation. Negative posts nominally lead, which would be a tempting "outrage drives engagement" headline, but the effect size does not support it and the correlation coefficient confirms as much. Reported as null.

6. Views, procedures, function — sql/08

Object	Purpose
vw_post_enriched	posts $\bowtie$ users, engagement rate, temporal parts, has_real_time flag
vw_analysis_ready	observed likes, real text, known platform — the population for any relationship claim
vw_data_quality	one-query scorecard of corpus state
usp_top_brands_by_platform	parameterised ranking — @top, @min_posts, @observed_only
usp_anomaly_report	anomaly counts for an arbitrary date window, derived not cached
fn_platform_percentiles	inline TVF; the optimiser folds it into the caller rather than running it row by row

The screenshot shows a SQL Server Enterprise Manager interface. The query window displays the following SQL code:

```

1  /-=====*/
2  DATA VORTEX :: Round 1 Phase 2
3  08 ~ Views, stored procedures, table-valued function
4
5  The rulebook explicitly permits CTEs, window functions, views and
6  procedures. These are not decoration: each one removes a join or a
7  filter that would otherwise be retyped in every analysis query, which
8  is where quiet inconsistencies creep in.
9  ===== */

```

The Results pane shows a summary of corpus state:

metric	value
1 total posts	12000
2 total users	1500
3 posts with imputed likes	2323
4 posts with missing text	1779
5 posts with unknown platform	1794
6 contradictory sentiment	329
7 date-only timestamps	3526
8 fully clean posts	7012

The Results pane also shows a detailed view of top brands by platform:

platform	rank	brand	posts	avg_engagement
1 Facebook	1	Toyota	134	4291.85
2 Facebook	2	Pepsi	136	4178.81
3 Facebook	3	Apple	156	4163.38
4 Instagram	1	Pepsi	134	4214.67
5 Instagram	2	Coca-Cola	123	4188.53
6 Instagram	3	Nike	138	4158.99
7 Reddit	1	Samsung	145	4239.85
8 Reddit	2	Pepsi	145	3976.63

The Results pane also shows a detailed view of data quality metrics:

window_from	window_to	platform	posts	contradictory_sentiment	shares_exceed_likes	imputed_likes	missing_text	beyond_3_sigma
1 2024-05-01	2025-04-30	Facebook	2074	62	345	389	311	0
2 2024-05-01	2025-04-30	Instagram	1989	48	333	392	300	0
3 2024-05-01	2025-04-30	Reddit	2031	57	342	368	315	0
4 2024-05-01	2025-04-30	Twitter	2049	54	362	417	307	0
5 2024-05-01	2025-04-30	Unknown	1784	49	292	348	243	0
6 2024-05-01	2025-04-30	YouTube	2073	59	313	409	303	0

The Results pane also shows a detailed view of platform percentiles:

platform	p25	p50	p75	p95
1 Reddit	2064	4011	4998.5	6478.5

The status bar at the bottom indicates: Query executed successfully. SAHIL\SQLEXPRESS (16.0 RTM) SAHIL\ntati (59) DataVortex 00:00:01 35 rows

vw_analysis_ready is the one that earns its place: it encodes the three filters every relationship claim in section 5 depends on, so the rule cannot be forgotten in a later query. has_real_time does the same job for the Q4.4 filter.

7. Top findings

#	Finding	Query	Why it matters
1	Corruption was applied per-column, not per-record ($\chi^2 = 2.295$, $p = 0.130$)	Q5.3	Dropping incomplete rows would discard ~30% of the corpus for nothing
2	The midnight spike is an ingest artefact — 3,526 date-only rows	Q4.4	Invalidates any time-of-day claim made without ts_source_format
3	Zero posts beyond 3σ ; $\max z = 2.693$	Q5.1	An IQR-only anomaly section finds nothing; the real anomalies are structural and semantic
4	329 posts (2.74%) carry contradictory sentiment, engaging no differently from clean posts	Q5.2	Invisible to every metric-based detector; only text analysis finds it
5	Engagement metrics are mutually independent ($ r < 0.03$) and independent of follower count ($r = 0.001$)	5.1, 5.2	No organic process links them; constrains what any model on this data could claim
6	Imputation contracts $\sigma(\text{likes})$ by 10.1% while leaving the mean unchanged	5.4	Justifies carrying likes_imputed from Phase 1 into the database
7	A $457\times$ spread in followers produces a 0.7% spread in engagement	Q6.1	The clearest single refutation that this corpus encodes real social behaviour
8	Hashtag vocabulary is 29 tags, not 56	Phase 1 D5b	27 phantom tags from a mis-ordered mojibake repair would have corrupted Q6.5 and Q4.3 entirely

The through-line: on a dataset this uniform, the discipline that matters is refusing to report effects the effect sizes do not support. Six of the eight findings above are constraints on what can be claimed, not claims.

8. Limitations

- 2,323 likes values (19.4%) are group medians, not observations. Every relationship claim excludes them; volume and coverage claims do not.

- 3,526 timestamps carry no time component. Time-of-day analysis covers the remaining 8,474 rows only.
- 1,784 posts have no known platform. Retained as Unknown rather than inferred or dropped, so platform-level figures exclude a known 14.9% of the corpus.
- 1,779 posts have no text, so text-derived features — sentiment, brand, hashtags, contradiction — are null for them. The contradiction rate of 2.74% is against all 12,000 posts, not against the scoreable subset.
- Engagement metrics show near-uniform distributions and near-zero mutual correlation throughout, consistent with synthetic generation. Findings about *relationships* between metrics should be read as measurements of that generator, not of human behaviour.
- Sentiment is lexicon-based and anchored on the verdict clause. This is exact for a template corpus with closed vocabularies; it would not transfer to free-form text.

Repository: <https://github.com/Sahil-Patel180/datavortex26> **Phase 1 artefacts:**

round_1/data/processed/, round_1/docs/cleaning_decisions.md **Phase 2 scripts:**

round_1/sql/00_create_db.sql ... 08_views_procs.sql